Regression Models — Cheatsheet

Equations & Key Properties

1. Two-Sample t-Test

Setup: $Y_s \sim N(\mu_1, \sigma^2)$, $Y_{ns} \sim N(\mu_2, \sigma^2)$, equal variance assumed.

$H_0 : \mu_1 = \mu_2$ vs $H_1 : \mu_1 \neq \mu_2$

Pooled variance:

$$S_p^2 = \frac{(n_s - 1)\, S_s^2 + (n_{ns} - 1)\, S_{ns}^2}{n_s + n_{ns} - 2}$$

Test statistic:

$$T = \frac{\hat{\mu}_s - \hat{\mu}_{ns}}{S_p \sqrt{\dfrac{1}{n_s} + \dfrac{1}{n_{ns}}}} \sim t_{n_s + n_{ns} - 2}$$

Reject H_0 if $|T| > t_{\alpha/2,\, df}$. **p-value:** $p = 2\, P(T > |t|)$.

95% Confidence Interval:

$$(\hat{\mu}_s - \hat{\mu}_{ns}) \pm t_{.025,\, df}\; S_p \sqrt{\frac{1}{n_s} + \frac{1}{n_{ns}}}$$

SBP numerical result: $T = -2.52$, $p = 0.019$, CI $= (-10.59, -1.05)$
$\Rightarrow$ Significant at 5% level — reject H_0.

2. Simple Linear Regression

Data: $\{(y_i, x_i),\ i = 1, \ldots, n\}$. Y = response, x = predictor.

Model:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\mu_i = E(Y \mid x_i) = \beta_0 + \beta_1 x_i, \qquad \mathrm{Var}(Y \mid x_i) = \sigma^2$$

Error properties: ε_i independent, $E(\varepsilon_i \mid x_i) = 0$, $\mathrm{Var}(\varepsilon_i \mid x_i) = \sigma^2$, $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

β_0: mean response at $x = 0$ (intercept).
$\beta_1 = E(Y \mid x_i + 1) - E(Y \mid x_i)$ (slope — change in mean response per unit change in x).

3. OLS Estimation

Minimize the Error Sum of Squares:

$$\mathrm{ESS} = \sum_{i=1}^{n} \big(Y_i - \beta_0 - \beta_1 x_i\big)^2$$

MLE $\equiv$ OLS under normality of errors.

Score equations:

$$\frac{\partial\, \mathrm{ESS}}{\partial \beta_0} = 0, \qquad \frac{\partial\, \mathrm{ESS}}{\partial \beta_1} = 0$$

OLS Solutions:

$$\boxed{\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum x_i Y_i - n \bar{x}\, \bar{Y}}{\sum x_i^2 - n \bar{x}^2}}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\, \bar{x} \qquad \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1\, x$$

Sums of squares notation:

$$S_{xx} = \sum (x_i - \bar{x})^2, \quad S_{yy} = \sum (Y_i - \bar{Y})^2, \quad S_{xy} = \sum (x_i - \bar{x})(Y_i - \bar{Y})$$

4. Sampling Distributions

$\hat{\beta}_1 = \sum w_i Y_i$, where $w_i = \dfrac{x_i - \bar{x}}{S_{xx}}$, $\sum w_i = 0$ (linear combination of Y_i).

$$\hat{\beta}_1 \sim N\left(\beta_1,\ \frac{\sigma^2}{S_{xx}}\right), \qquad E(\hat{\beta}_1) = \beta_1$$

$$\hat{\beta}_0 \sim N\left(\beta_0,\ \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]\right), \qquad E(\hat{\beta}_0) = \beta_0$$

5. Error Variance Estimation

Residual: $\hat{\varepsilon}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$

ANOVA decomposition:

$$S_{yy} = \underbrace{\sum \hat{\varepsilon}_i^2}_{SS_R\ \text{(Residual)}} + \underbrace{\hat{\beta}_1^2\, S_{xx}}_{SS_M\ \text{(Model)}} \qquad \text{df: } (n-1) = (n-2) + 1$$

$$SS_R = \frac{S_{yy}\, S_{xx} - S_{xy}^2}{S_{xx}}, \qquad \boxed{\hat{\sigma}^2 = \frac{SS_R}{n-2} = MS_R}, \qquad \frac{SS_R}{\sigma^2} \sim \chi^2_{n-2}$$

6. Inference on β_1 (Slope)

$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_1 : \beta_1 \neq 0$

$$T = \frac{\hat{\beta}_1}{\hat{\sigma}/\sqrt{S_{xx}}} \sim t_{n-2}$$

Reject H_0 if $|T| > t_{n-2,\,\alpha/2}$. $\quad p = 2\,P(T > |t|)$.

95% CI for β_1:

$$\hat{\beta}_1 \pm t_{n-2,\,.025}\ \frac{\hat{\sigma}}{\sqrt{S_{xx}}}$$

Weight → SBP numerical result: $\hat{\beta}_1 = 0.416$, SE $= 0.083$, $T = 5.031$
95% CI $= (0.242,\ 0.590)$ — slope is significant.

7. Inference on Mean Response μ_{x_0}

$$\hat{\mu}_{x_0} = \hat{\beta}_0 + \hat{\beta}_1\, x_0 \sim N\big(\mu_{x_0},\ \mathrm{Var}(\hat{\mu}_{x_0})\big)$$

$$\mathrm{Var}(\hat{\mu}_{x_0}) = \sigma^2\left[\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}}\right]$$

Confidence Interval:

$$\hat{\mu}_{x_0} \pm t_{n-2,\,\alpha/2}\ \hat{\sigma}\ \sqrt{\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}}}$$

Narrowest at $x_0 = \bar{x}$; widens as $|x_0 - \bar{x}|$ increases.

8. Coefficient of Determination R^2

$SS_T = S_{yy}, \quad SS_R = \sum(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2, \quad SS_M = \hat{\beta}_1^2 \, S_{xx}$

$$\boxed{R^2 = 1 - \frac{SS_R}{SS_T} = \frac{S_{xy}^2}{S_{xx}\, S_{yy}}}$$

$0 \le R^2 \le 1$. $100\, R^2\%$ of the variation in Y is explained by the linear relationship with x.

Weight → SBP: $SS_T = 3409.2$, $SS_R = 1417.0$, $R^2 = 0.584$
$\Rightarrow$ 58.4% of the variation in SBP is explained by weight.

9. Model Diagnostics

Residuals vs. Fitted plot — checks independence, constant variance, and zero mean of residuals.

Q–Q plot — checks normality of ε_i. Use $\hat{\varepsilon}_i/\hat{\sigma}$ (standardized residuals).

Pattern	Interpretation
Random scatter	Model assumptions satisfied
Fan shape	Heteroscedasticity (non-constant variance)
Curved pattern	Non-linearity — consider transformation

10. Key Numerical Results

Problem	Fitted Model	Key Statistic
Smoking → SBP	$\hat{Y} = 123.36 + 5.825\, x$	$\hat{\beta}_1 = 5.825$
Weight → SBP	$\hat{Y} = 68.58 + 0.416\, x$	$R^2 = 0.584$
Mother → Daughter	$\hat{Y} = 46.59 + 0.315\, x$	$\hat{\beta}_1 = 0.315$

11. Matrix Notation (Simple LR)

$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$:

$$\underbrace{\begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}}_{\mathbf{Y}} = \underbrace{\begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}}_{\mathbf{X}} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}}_{\boldsymbol{\beta}} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}}_{\boldsymbol{\varepsilon}}$$

$$E(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n, \quad E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$$

12. OLS in Matrix Form

$$\mathrm{ESS} = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$$

$$\mathbf{X}'\mathbf{X}\,\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y} \quad \text{(normal equations)}$$

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}}$$

Simple LR special case:

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{n\, S_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}$$

13. Properties of $\hat{\boldsymbol{\beta}}$

$$E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} \quad \text{(unbiased)}$$

$$\boxed{\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2\,(\mathbf{X}'\mathbf{X})^{-1}}$$

Gauss–Markov Theorem: Under the classical linear model assumptions, the OLS estimator $\hat{\boldsymbol{\beta}}$ is **BLUE** — Best Linear Unbiased Estimator. It has the smallest variance among all linear unbiased estimators.

14. Multiple Linear Regression

Model:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \varepsilon_i \qquad \Longleftrightarrow \qquad \mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{p1} \\ 1 & x_{12} & \cdots & x_{p2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & \cdots & x_{pn} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$$

$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}, \quad E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}, \quad \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$

Interpretation: β_j = change in $E(Y)$ for a 1-unit increase in x_j, holding all other predictors constant.

$$\beta_j = E(Y \mid x_j + 1, \mathbf{x}_{-j}) - E(Y \mid x_j, \mathbf{x}_{-j})$$

$\beta_0 = E(Y \mid x_1 = 0, \ldots, x_p = 0)$ (intercept).

Suicide rate model example: $\hat{Y} = 3.507 - 0.0002\, x_1 + 0.261\, x_2$
x_1 = population (1000s), x_2 = divorce rate (per 100K). Only x_2 marginally significant ($p = 0.161$).

15. Quick Reference

Quantity	Formula
S_{xx}	$\sum_{i=1}^{n}(x_i - \bar{x})^2$
S_{yy}	$\sum_{i=1}^{n}(Y_i - \bar{Y})^2$
S_{xy}	$\sum_{i=1}^{n}(x_i - \bar{x})(Y_i - \bar{Y})$
$\hat{\beta}_1$	S_{xy}/S_{xx}
$\hat{\beta}_0$	$\bar{Y} - \hat{\beta}_1 \bar{x}$
$\hat{\sigma}^2$	$SS_R/(n-2)$
SS_R	$(S_{yy}\, S_{xx} - S_{xy}^2)/S_{xx}$
R^2	$S_{xy}^2/(S_{xx}\, S_{yy})$